

MyDesign Portfolio

Element C

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The socket organizer must only fit one socket in width and be able to fit 10 lengthwise.

The creation of this new organization system is better for productivity and we can see this in the article "Are Transformed Workplaces More Productively Efficient?" by David Fairris states that people become more productively efficient when an innovation in organization is made.

Anonio likes this as this will fit around the workshop better and can be moved better than leaving it fixed in one spot. He wants two of these rows to be constructed to fit 20 sockets that he needs.

Design Requirement #2 (HIGH Priority)

Easy to move around without having problems with the sockets falling off the organizer. (It should have the capability to be moved to the location that you are using the tools)

This makes it so that when moving around the shop, problems where sockets could get lost or roll away would not happen as they would be secure to the holder. The less movement an employee must make decreases the time it takes to complete an individual task as stated by Amanda Hamel in her analysis of workplace efficiency

Design Requirement #3 (HIGH Priority)

The socket organizer must be made of durable materials because it needs long term use in the workshop.

The socket organizer should be made with high quality materials because it is desirable for the users of the organizer to be able to use it for a long time and hold up in an automotive shop environment



Design Requirement #4 (HIGH Priority)

Able to mount on and be removed from a standard pegboard with little effort and in a timely fashion. This would allow the tool to be modular if the shop were to change the location and size of the tool area / pegboard

Our stakeholder will be storing the sockets on his pegboard, so we need to make it able to hook onto the pegboard

Design Requirement #6 (LOW Priority)

Antonio would prefer the pegboard to be a simple black or white for aesthetic purposes to match with the other tools and racks in the area. Most of Antonio's workshop matches the simple monochromatic black or white design. This design is aesthetic, but may serve to increase employee satisfaction in the visual appeal of their environment which may increase efficiency. As said by Zhenjing the better the employees are satisfied with the work environment the better they work and since Antonio is the main worker if it matches what he likes aesthetically he will work better.

Design Requirement #7 (LOW Priority)

Labels for each socket size as this will allow workers to more easily find the right size of socket that they need to use, this will also increase the speed at which workers are able to find the required sockets